

of living conditions are infant mortality rate and life expectancy (see Concept 53.6). From 1930 to 2010, the infant mortality rate in Costa Rica declined from 170 to 9 per 1,000 live births; over the same period, life expectancy increased from about 43 years to 79 years. Another indicator of living conditions is the literacy rate. The 2011 literacy rate in Costa Rica was 96%, compared to an average of 82% in the other six Central American countries. Such statistics show that living conditions in Costa Rica have improved greatly in the time in which the country has dedicated itself to conservation and restoration. While this does not prove that conservation *causes* an improvement in human welfare, we can say with certainty that development in Costa Rica has attended to both nature *and* people.

▼ **Figure 56.36** Biophilia, past and present.



(a) Detail of animals in a 17,000-year-old cave painting, Lascaux, France



(b) A 30,000-year-old ivory carving of a water bird, found in Germany



(c) Nature lovers on a wildlife-watching expedition

The Future of the Biosphere

Our modern lives are very different from those of early humans, who hunted and gathered to survive. Their reverence for nature is evident in the early cave paintings of wildlife (Figure 56.36a) and in the visions of life they sculpted from bone and ivory (Figure 56.36b).

Our lives reflect remnants of our ancestral attachment to nature and the diversity of life—the concept of *biophilia* introduced early in this chapter. We evolved in natural environments rich in biodiversity, and we still have an affinity for such settings (Figure 56.36c and d). Indeed, our biophilia may be innate, an evolutionary product of natural selection acting on a brainy species whose survival depended on a close connection to the environment and a practical appreciation of plants and animals.

Our appreciation of life guides the field of biology today. We celebrate life by deciphering the genetic code that makes each species unique. We embrace life by using fossils and DNA to chronicle evolution through time. We preserve life through our efforts to classify and protect the millions of species on Earth. We respect life by using nature responsibly and reverently to improve human welfare.

Biology is the scientific expression of our desire to know nature. We are most likely to protect what we appreciate and are most likely to appreciate what we understand. By learning about the processes and diversity of life, we become more aware of our place in the biosphere. We hope this text has served you well in this lifelong adventure.

CONCEPT CHECK 56.5

1. What is meant by the term *sustainable development*?
2. How might biophilia influence us to conserve species and restore ecosystems?
3. **WHAT IF?** Suppose a new fishery is discovered, and you are put in charge of developing it sustainably. What ecological data might you want on the fish population? What criteria would you apply for the fishery's development?

For suggested answers, see Appendix A.

(d) A young biologist holding a songbird

